

Mini-Project Deliverable — Problem Framing Memo

Project Title

AI-Based Alloy Manufacturing Analytics System

Problem Statement

The alloy ingot manufacturing process faces challenges related to inconsistent alloy purity and excessive waste material generation. Variations in furnace conditions, raw material quality, machine performance, and environmental factors can negatively affect production quality and operational efficiency. The objective of this project is to use Artificial Intelligence and Machine Learning techniques to improve manufacturing quality and reduce operational losses.

Target Definition

- purity_pass → Classification
- purity_percentage → Regression
- waste_material_kg → Regression

Safe Feature List

- furnace_temperature
- furnace_pressure
- energy_consumption
- impurity_level
- humidity_level
- machine_vibration
- slag_percentage
- maintenance_frequency
- downtime_minutes
- recycling_rate
- cooling_rate
- melting_time
- batch_size_kg
- raw_material_quality
- chemical_composition
- machine_condition
- operator_shift
- operator_experience_years
- power_usage_kwh
- IoT_sensor_data

Leakage Exclusions

- final_lab_result
- rejected_batch_status

- customer_feedback
- post_inspection_notes
- final_production_report

Proposed Train / Validation / Test Split

- Training Set: 70%
- Validation Set: 15%
- Test Set: 15%

Success Metrics

Classification Metrics:

- Accuracy
- Precision
- Recall
- F1-Score

Regression Metrics:

- Mean Absolute Error (MAE)
- Root Mean Squared Error (RMSE)
- R² Score

Top Three Risks and Mitigations

1. Data Leakage → Remove all post-production variables.
2. Poor Sensor Quality → Perform regular sensor calibration and maintenance.
3. Imbalanced Production Data → Use balanced sampling and continuous data collection.

Conclusion

This project demonstrates how AI and machine learning can support alloy manufacturing through predictive analytics, quality control, waste reduction, and smart manufacturing concepts.